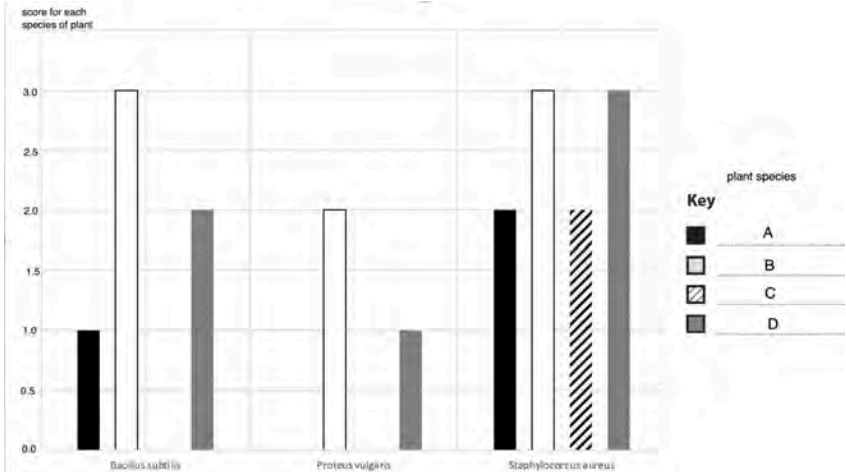


Question Number	Answer	Additional Guidance	Mark
1a	An answer that includes five of the following points. • description of {bacteria grown (on nutrient agar) / bacterial lawn} / bacterial broth (1) • filter paper discs soaked in plant extract (placed on agar) / plant extract placed in well (cut in agar) / plant extract added to broth (1) • control variable described (1) • one safety aspect described (1) • incubate for suitable time (1) • clear zone around {disc / well} {measured / scored / turbidity measured (1)}	Accept microbe e.g. disc size / extract volume / temperature e.g. flaming / dish lid / use autoclave / safe temperature quoted (<32 C) Accept 24 hours to 1 week Accept cloudiness	(5)

Question Number	Answer	Additional Guidance	Mark																				
1(b)(i)	A chart with the following features. - Y-axis correctly completed (1) - X-axis correctly completed (1) - key correctly completed (1)	correct linear scale and title bacteria names A,B,C,D and title  <table border="1"><caption>Score for each species of plant</caption><thead><tr><th>Plant Species</th><th>A</th><th>B</th><th>C</th><th>D</th></tr></thead><tbody><tr><td>Bacillus subtilis</td><td>1.0</td><td>3.0</td><td>0.0</td><td>2.0</td></tr><tr><td>Proteus vulgaris</td><td>0.0</td><td>2.0</td><td>0.0</td><td>1.0</td></tr><tr><td>Staphylococcus aureus</td><td>2.0</td><td>3.0</td><td>2.0</td><td>3.0</td></tr></tbody></table>	Plant Species	A	B	C	D	Bacillus subtilis	1.0	3.0	0.0	2.0	Proteus vulgaris	0.0	2.0	0.0	1.0	Staphylococcus aureus	2.0	3.0	2.0	3.0	(3)
Plant Species	A	B	C	D																			
Bacillus subtilis	1.0	3.0	0.0	2.0																			
Proteus vulgaris	0.0	2.0	0.0	1.0																			
Staphylococcus aureus	2.0	3.0	2.0	3.0																			

Question Number	Answer	Additional Guidance	Mark
1(b)(ii)	An answer that includes the following points: • plant C has the lowest (antimicrobial) effect / affects only one bacterial species (1) • plant B has the highest (antimicrobial) effect (1) • {B / D} affect all 3 species (1) • data manipulated to support one of the first two marking points (1)	e.g. B has a total score of 8 / C has a total score of 2	(4)